

# ConceptGrade: A Knowledge Graph–Driven Framework for Concept-Aware Automated Short Answer Grading in Computer Science Education

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**Abstract**—Automated Short Answer Grading (ASAG) in Computer Science (CS) education remains a challenging open problem due to the technical nature of CS knowledge, the diversity of valid responses, and the need to assess conceptual correctness rather than surface-level textual similarity. Existing approaches—based on lexical overlap, embedding similarity, or zero-shot large language models (LLMs)—lack explicit representations of domain knowledge and cannot identify which specific concepts a student understands or misses. We present ConceptGrade, a five-layer concept-aware ASAG framework that transforms student responses into *concept graphs* and compares them against expert-curated domain knowledge graphs (KGs). The system extracts concepts and relationships from free-text answers using a chain-of-thought LLM pipeline, computes semantic coverage and structural integration scores via graph matching, and synthesizes a composite grade through a weighted scoring model. We construct an expert Data Structures KG comprising 101 domain concepts and 138 typed relationships spanning eight semantic categories. We evaluate ConceptGrade on  $n = 1,239$  graded student answers spanning 177 questions across three independent ASAG datasets: Mohler 2011 (CS Data Structures,  $n = 120$ , 10 questions), DigiKlausur (Neural Networks,  $n = 646$ , 17 questions), and Kaggle ASAG (Elementary Science,  $n = 473$ , 150 questions). On the in-domain Mohler benchmark, ConceptGrade achieves Pearson  $r = 0.982$ , QWK = 0.975, and MAE = 0.223, a 32.4% MAE reduction over the identical-model LLM zero-shot baseline (Wilcoxon two-tailed  $p = 0.0026$ ; one-tailed  $p = 0.0013$ ; paired  $d_z = 0.30$ ; bootstrap 95% CI for the MAE reduction on the  $n = 90$  test split: [15.1%, 49.7%]). A signal-source ablation shows that this reduction is carried by the high-reliability KG concept-coverage signal: a **concepts\_only** ablation, which omits the misconception taxonomy (machine-IRR  $\kappa = 0.33$ ) and does not consume any output of the Mohler-augmented Verifier fine-tune, achieves a slightly better MAE of 0.217 (34.2% reduction). The headline result is therefore not reliant on the fair-IRR taxonomy or on Verifier-fine-tune transfer. Across all three datasets the effect is heterogeneous ( $I^2 = 70\%$ , substantial heterogeneity in the Higgins–Thompson sense), so we report the per-dataset results as the primary cross-dataset evidence rather than a single pooled number: ConceptGrade improves over the baseline on Mohler ( $d_z = -0.30$ ,  $p_{\text{two}} = 0.003$ ), shows a small but significant improvement on DigiKlausur ( $d_z = -0.07$ ,  $p_{\text{one}} = 0.024$ , practical effect below Cohen’s small-effect threshold), and shows no significant improvement on Kaggle ASAG ( $d_z = 0.02$ ,  $p_{\text{one}} = 0.60$  clustered, or  $p_{\text{one}} = 0.17$  at response level). On Kaggle ASAG

the LLM extraction module produced empty matched-concept sets for 473/473 answers, an upstream failure consistent with a deliberate-architectural prediction: the KG-grounding effect requires a non-empty KG–answer intersection, and elementary-science colloquial language does not provide one. We treat the three-dataset behaviour as a *boundary characterisation*, not a generalisation claim. A six-condition ablation confirms that concept coverage is the most influential component ( $\Delta\text{QWK} = -0.416$  when removed). These results demonstrate that KG-aware grading provides substantial gains over the identical-model LLM baseline in vocabulary-specific domains and characterise the boundary conditions beyond which the effect attenuates.

**Index Terms**—automated short answer grading, knowledge graph, concept extraction, computer science education, large language models, Bloom’s taxonomy, NLP

## I. INTRODUCTION

**Scope of this paper (what we do not claim).** We do not claim state-of-the-art performance on the full Mohler 2011 benchmark or methodological novelty in any single layer of the pipeline. Our empirical claims are confined to a pooled  $n = 1,239$  paired predictions across three independent ASAG datasets and 177 questions (Section ??): (i) a 32.4% paired-MAE reduction on the in-domain Mohler 2011 KG-aligned subset against the identical zero-shot LLM ( $d_z = -0.30$ , one-tailed clustered  $p < 0.05$  in all 10 LOOCV folds); (ii) a small but significant cross-dataset pooled effect under fixed-effects meta-analysis ( $d_z = -0.073$ ,  $p_{\text{two}} = 0.010$ ) with substantial between-dataset heterogeneity ( $I^2 = 70\%$ ) that itself maps to domain vocabulary specificity; (iii) boundary behaviour on DigiKlausur (marginally significant: response-level  $p_{\text{two}} = 0.049$ ; clustered  $p$  above 0.05) and Kaggle ASAG (not significant at either response or cluster level); and (iv) an engineering integration of LLM concept extraction, KG matching, and confidence-weighted synthesis whose component-level effect on accuracy is quantified by a six-row ablation. Comparison to frontier LLMs and retrieval-augmented baselines, and replication on an independently authored KG, are explicit future-work items (§??). A reader looking for state-

of-the-art on the full benchmark or methodological novelty in a single layer should regard those as out of scope.

Providing timely, consistent, and educationally meaningful feedback on student short answers is a critical pedagogical task in CS courses. Instructors of introductory data structures or algorithms classes routinely evaluate hundreds of written responses per assignment, each requiring domain-specific judgment about conceptual correctness, completeness, and depth. Manual grading at this scale is labor-intensive, subject to inter-rater variability, and delays feedback to the point where students can no longer act on it.

Automated Short Answer Grading (ASAG) systems aim to replicate expert grading behavior. However, the CS domain presents unique challenges: (1) a student can correctly define a stack without mentioning “LIFO,” rendering bag-of-words approaches unreliable; (2) two answers may use entirely different vocabulary yet reflect equivalent understanding; (3) a technically correct sentence embedded in a response full of misconceptions should not receive full credit; and (4) the *structure* of knowledge—whether concepts are merely listed or genuinely integrated—reflects qualitatively different levels of understanding.

Contemporary ASAG approaches fall into three broad families.

*Lexical approaches* [1] use TF-IDF or n-gram overlap; they are fast but blind to semantic equivalence.

*Semantic similarity approaches* [3] use sentence-level alignment and paraphrase features; Sultan et al. (2016) achieved  $r = 0.592$  without neural representations. *Embedding approaches* [6] subsequently apply pre-trained transformers such as BERT [5] to encode answer pairs, reaching  $r = 0.620$  on the Mohler benchmark through task-specific fine-tuning on SemEval training data. While BERT-fine-tuned systems represent the strongest prior non-LLM baseline, they still lack explicit domain knowledge and require substantial labeled fine-tuning data not available for our KG-aligned evaluation subset.

*LLM zero-shot approaches* prompt large models like GPT-4, Gemini, or Llama to produce grades directly [14], [15]. While surprisingly capable—our zero-shot baseline (gemini-2.5-flash, see below) achieves  $r = 0.971$  on the 10-question KG-aligned subset—they offer no explicit conceptual representation, are inconsistent across runs, and cannot enumerate which concepts are missing. ConceptGrade uses the LLM as one structured component (concept extraction) rather than an end-to-end oracle. We deliberately compare against *the same gemini-2.5-flash model* as the zero-shot baseline rather than a frontier model (GPT-4-turbo, Claude 3.5 Sonnet) so that the comparison isolates the architectural contribution (KG-grounding + multi-layer synthesis) from model capability: C\_LLM and C5\_fix use the same model, same temperature ( $T = 0$  via Gemini batch API), and the same scoring rubric, differing only in whether the prompt receives KG-grounding evidence. Comparison against frontier and retrieval-augmented baselines is reported as a controlled-LLM design choice rather than an oversight, with the appropriate extension left to future work.

None of these approaches answer the questions an instructor most needs: *Which concepts did the student demonstrate? Which are missing from their response? How deeply integrated is their understanding?*

This paper presents **ConceptGrade**, a knowledge graph-driven framework that addresses these limitations. Our core insight is that grading a CS short answer can be framed as a *knowledge graph comparison problem*: transform the student’s response into a concept graph and measure how well it matches an expert-curated domain KG. This enables not only a score, but also fine-grained diagnostic feedback—a prerequisite for actionable, personalized learning support.

**Contributions.** This paper makes the following contributions:

- 1) A five-layer ASAG architecture combining concept extraction, KG comparison, cognitive depth assessment, misconception detection, and score synthesis.
- 2) An expert Data Structures domain KG with 101 concepts and 138 typed relationships, manually curated following CS education ontology principles.
- 3) A hybrid LLM + rule-based concept extraction pipeline that produces structured `StudentConceptGraph` objects from free-text answers.
- 4) A KG comparison module computing coverage, integration, and accuracy scores via structural graph matching.
- 5) An empirical evaluation on the Mohler et al. (2011) benchmark with bootstrap confidence intervals and Wilcoxon significance testing.
- 6) A full ablation study quantifying the contribution of each framework component.

## II. RELATED WORK

### A. Automated Short Answer Grading

ASAG research spans three decades, evolving from hand-crafted rubrics to neural systems. Early work by Mohler and Mihalcea [1] established the benchmark using Latent Semantic Analysis (LSA) and WordNet-based measures, achieving Pearson  $r = 0.493$  on their CS dataset. Mohler et al. [2] extended this with dependency graph alignment ( $r = 0.518$ ). Sultan et al. [3] introduced sentence-level semantic similarity features ( $r = 0.592$ ). More recently, transformer-based fine-tuning on SemEval datasets [4] has become the dominant paradigm, with BERT-based systems reaching  $r = 0.620$  on the Mohler set.

Despite these advances, few prior ASAG systems for CS explicitly construct a domain knowledge graph from the student’s free-text response and use graph-structural comparison as the primary grading signal. Most approaches treat the task as regression over sentence-pair similarity—a formulation that conflates syntactic fluency with conceptual accuracy and cannot identify which specific concepts are present, absent, or incorrectly related.

### B. Knowledge Graphs in Education

Knowledge graphs have been applied to intelligent tutoring systems (ITS) [11], prerequisite learning [12], and concept map assessment [13]. Concept map scoring measures the structural

similarity between student-drawn and expert concept maps. ConceptGrade extends this idea to *free-text* responses: we automatically extract the concept map from prose rather than requiring the student to draw it explicitly, making the approach practical at classroom scale.

### C. LLM-Based Assessment

Recent work has explored zero-shot or few-shot LLM prompting for ASAG [14], [15]. While results are promising, LLM-only systems face issues of inconsistency, hallucinated justifications, and lack of transparency. ConceptGrade uses an LLM as one component (concept extraction) within a structured pipeline, preserving explainability while benefiting from language understanding capabilities.

### D. Bloom’s Taxonomy in Automated Assessment

Emirtekin and Özarslan [7] demonstrate LLM-based Bloom’s taxonomy classification achieving QWK = 0.585–0.640. ConceptGrade integrates a Bloom’s classifier as Layer 3 but treats it as a component of a broader scoring framework rather than as the primary assessment signal.

## III. METHODOLOGY

### A. System Overview

ConceptGrade processes each student response through five sequential layers, illustrated in Fig. 1. Layers 1–2 constitute the knowledge-aware grading core and are the focus of this paper. Layers 3–5 (cognitive depth, misconception detection, and score synthesis) leverage the Layer 1–2 outputs and are described in §III-G.

### B. Expert Domain Knowledge Graph Construction

The Data Structures domain KG is the ground-truth representation of expert knowledge against which student responses are evaluated. We manually curated it following three design principles: (1) *pedagogical coverage*—all concepts typically taught in a first or second-year CS data structures course are included; (2) *semantic precision*—each concept has a unique canonical identifier, description, and difficulty level (1–4); (3) *relational richness*—relationships carry explicit semantic types rather than being treated as generic edges.

**Concepts.** The KG contains 101 concepts across eight semantic types. Table I summarizes the concept type distribution. Concepts span four difficulty levels: Level 1 (foundational, e.g., array, stack), Level 2 (core, e.g., binary\_search\_tree, hash\_table), Level 3 (advanced, e.g., avl\_tree, dijkstra), and Level 4 (expert, e.g., red\_black\_tree, b\_tree).

**Relationships.** The KG contains 138 typed relationships across 11 relation types. Typed relationships enable semantically grounded comparison: a student who says “a heap is a tree” is expressing an *is\_a* relationship, while “heap sort uses a heap” expresses *uses*. Table II shows the relationship type distribution. Every edge carries a weight in [0, 1] reflecting relationship strength.

figures/fig1\_architecture.png

Fig. 1. ConceptGrade five-layer architecture. Student answers are transformed into concept graphs (Layer 1), compared against an expert domain KG (Layer 2), assessed for cognitive depth via Bloom’s and SOLO classifiers (Layer 3), checked for misconceptions (Layer 4), and synthesized into a composite score (Layer 5).

TABLE I  
DATA STRUCTURES KNOWLEDGE GRAPH: CONCEPT TYPE DISTRIBUTION

| Concept Type     | Count      | Examples                                 |
|------------------|------------|------------------------------------------|
| Algorithm        | 22         | merge_sort, bfs, dijkstra                |
| Data Structure   | 18         | stack, queue, binary_tree                |
| Abstract Concept | 15         | recursion, pointer, iteration            |
| Property         | 14         | lifo, fifo, balanced, stable             |
| Operation        | 12         | insertion, deletion, traversal           |
| Complexity Class | 8          | $O(1)$ , $O(\log n)$ , $O(n)$ , $O(n^2)$ |
| Prog. Construct  | 7          | array, reference, memory                 |
| Design Pattern   | 5          | divide_conquer, chaining                 |
| <b>Total</b>     | <b>101</b> |                                          |

### C. Concept Extraction (Layer 1)

Given a question  $q$  and student answer  $a$ , Layer 1 constructs a StudentConceptGraph  $G_s = (V_s, E_s)$  where nodes  $V_s$  represent extracted concepts and edges  $E_s$  represent relationships.

We employ a chain-of-thought (CoT) LLM prompt that includes: (1) the full domain KG as context (flattened,  $\approx 5,000$  tokens); (2) the question and student answer; (3) structured instructions to extract concept IDs, relationship types, correctness flags, and a depth assessment. The LLM (gemini-2.5-flash via the Google Generative AI batch API) returns structured JSON:

```
{
  "concepts": [
    { "concept_id": "stack",
      "confidence": 0.95,
```

TABLE II  
DATA STRUCTURES KNOWLEDGE GRAPH: RELATIONSHIP TYPES

| Relation Type    | Semantics                | Count      |
|------------------|--------------------------|------------|
| has_property     | Attribute assignment     | 25         |
| is_a             | Taxonomic hierarchy      | 22         |
| uses             | Algorithmic dependency   | 20         |
| has_complexity   | Complexity class binding | 18         |
| prerequisite_for | Learning dependency      | 15         |
| variant_of       | Structural similarity    | 14         |
| implements       | Concrete realization     | 12         |
| contrasts_with   | Conceptual opposition    | 8          |
| has_part         | Structural composition   | 4          |
| <b>Total</b>     |                          | <b>138</b> |

```

    "surface_form": "stack",
    "is_correct": true},
    ...
  ],
  "relationships": [
    {
      "source_id": "stack",
      "target_id": "lifo",
      "type": "has_property",
      "confidence": 0.92,
      "is_correct": true},
    ...
  ],
  "overall_depth": "moderate"
}

```

**Confidence Filtering:** To reduce spurious concept extraction, we filter out concepts with LLM confidence scores below a threshold (default:  $\tau = 0.70$ ). This parameter was tuned on the development split (30% held-out from Mohler) to maximize QWK while minimizing false positive concepts. The filtering logic is: For each extracted concept  $v$  with confidence  $\text{conf}(v)$ , we keep  $v$  if and only if  $\text{conf}(v) \geq \tau$ . All relationships are then filtered: an edge  $(u, v)$  is retained only if both  $u$  and  $v$  survived the confidence filter. This ensures the concept graph remains internally consistent (no dangling edges).

Each extracted concept is normalized to its canonical KG identifier (e.g., “LIFO,” “last-in first-out,” and “last in first out” all map to `lifo`) using the `find_concept_by_alias()` method.

#### D. Knowledge Graph Comparison (Layer 2)

Layer 2 computes a `ComparisonResult` by matching  $G_s$  against the expert graph  $G_e = (V_e, E_e)$ .

##### Concept Coverage Score:

$$\text{cov}(G_s, G_e) = \frac{|V_s \cap V_e|}{|V_e|} \quad (1)$$

where  $V_e$  is optionally restricted to the question-relevant subgraph.

##### Relationship Accuracy Score:

$$\text{acc}(G_s) = \frac{|\{e \in E_s : \text{is\_correct}(e) = \top\}|}{|E_s|} \quad (2)$$

##### Integration Quality Score:

$$\text{int}(G_s) = \frac{|\{v \in V_s : \deg(v) \geq 1 \text{ in } G_s\}|}{|V_s|} \quad (3)$$

measuring what fraction of concepts are connected (non-isolated).

##### Overall Comparison Score:

$$\text{score}(G_s, G_e) = \alpha \text{cov} + \beta \text{acc} + \gamma \text{int} \quad (4)$$

with  $\alpha = 0.5$ ,  $\beta = 0.3$ ,  $\gamma = 0.2$  (set by cross-validation on the development split).

The comparator also produces interpretable outputs including `missing_concepts`, `extra_concepts`, `incorrect_relationships`, and `feedback_points`—actionable strings that can be returned directly to the student.

#### E. Cognitive Depth Assessment (Layer 3)

Layer 3 infers the cognitive level demonstrated by the student response using two complementary taxonomies.

**Bloom’s Taxonomy Classifier:** A hybrid rule-based + LLM classifier assigns one of six Bloom’s levels (Remember through Create) [8], [9]. The rule-based component detects level-indicative verb patterns (e.g., “define,” “compare,” “design”), while the LLM component performs chain-of-thought reasoning. Outputs are ensembled by confidence weighting.

**SOLO Taxonomy Classifier:** We introduce the first automated SOLO (Structure of the Observed Learning Outcome) classifier for free-text CS responses [10]. SOLO’s five levels (Prestructural through Extended Abstract) measure the *structural complexity* of understanding, complementing Bloom’s cognitive demand axis. Our hybrid classifier uses KG structural features—concept count (capacity dimension) and integration quality (relating operation dimension)—as rule-based signals, combined with LLM chain-of-thought reasoning.

The ensemble algorithm is:

$$L_{\text{final}} = \begin{cases} L_{\text{LLM}} & \text{if } |L_{\text{LLM}} - L_{\text{rule}}| \leq 1 \\ \lfloor (L_{\text{LLM}} + L_{\text{rule}})/2 + 0.5 \rfloor & \text{otherwise} \end{cases} \quad (5)$$

#### F. Misconception Detection (Layer 4)

Layer 4 cross-references the `StudentConceptGraph` against a CS misconception taxonomy containing 16 entries across 7 topic areas (e.g., DS-STACK-01: confusing LIFO with FIFO; DS-TREE-01: incorrect BST search complexity). The taxonomy was developed through expert review of common student errors observed in the Mohler dataset. Each entry specifies the misconception description, severity (CRITICAL/MODERATE/MINOR), the involved KG concepts, the canonical incorrect claim (used as a lexical anchor), and a remediation hint. A rule-based fallback uses concept-pair co-occurrence patterns to identify matches when the LLM is unavailable.

**Inter-coder reliability (machine-IRR pilot).** As a precursor to a full human-coder reliability study, we computed a machine-IRR pilot on the 120 Mohler KG-aligned answers using two independent automated coding methods: (i) a *KG-rule coder*, which assigns a taxonomy entry when the student’s concept graph overlaps the entry’s concept set and the average human

grade falls below 3.0/5; and (ii) a *lexical coder*, which assigns a taxonomy entry when the student’s answer text shares  $\geq 2$  stemmed content tokens with the entry’s canonical incorrect claim. Both coders use the same data and same taxonomy but different evidence channels (graph structure vs. text surface), so the resulting Cohen’s  $\kappa$  reflects construct-validity overlap. The pooled micro-averaged  $\kappa = 0.33$  and macro-averaged  $\kappa = 0.29$  across all 16 entries fall in the “fair agreement” range [?], with the highest per-entry agreement on DS-LINK-03 ( $\kappa = 0.57$ ) and DS-TREE-01 ( $\kappa = 0.56$ ). Sortin and complexity entries showed the lowest agreement (DS-SORT-02  $\kappa = 0.00$ , DS-COMP-01  $\kappa = 0.16$ ). The full per-entry breakdown and the computation script (`compute_taxonomy_kappa.py`) are included in the supplementary materials. We interpret these numbers as a lower bound on human-coder agreement (humans agree more with each other than two heuristics with no shared internal state agree); a confirmatory human second-coder pass remains future work.

### G. Topological Reasoning Mapping (TRM) Algorithm

ConceptGrade uses approximate subgraph matching (not NP-hard subgraph isomorphism) to align student concept graphs against the expert knowledge graph. The algorithm proceeds in three phases:

1) *Phase 1: Concept Alignment*: Given student graph  $G_s = (V_s, E_s)$  and expert graph  $G_e = (V_e, E_e)$ , we compute semantic similarity between each student concept and each expert concept:

$$\text{SimScore}(v_s, v_e) = \text{SemanticSimilarity}(v_s.\text{text}, v_e.\text{name}) \quad (6)$$

where `SemanticSimilarity` is computed via pretrained BERT embeddings (cosine distance). Each student concept is matched to the expert concept with highest similarity score above 0.70.

2) *Phase 2: Relationship Matching*: For each edge  $(u_s, v_s) \in E_s$ , we find the best matching edge in  $G_e$ :

$$\text{EdgeMatch}(e_s, e_e) = \begin{cases} 1.0 & \text{type match \& Sim}(u_s, u_e) > 0.75 \\ 0.5 & \text{type match (node mismatch)} \\ 0.0 & \text{otherwise} \end{cases} \quad (7)$$

3) *Phase 3: Verifier Confidence Weighting*: **Important caveat — Verifier training data.** The LLM Verifier used in this Phase is fine-tuned on an augmented expansion of the 630-sample full Mohler dataset (Paper 2 §A.2). Because the Mohler test split used for evaluation in §?? is drawn from the same source dataset, the headline 32.4% MAE reduction has a potential data-leakage interpretation, addressed empirically in §VI via the `concepts_only` ablation (which omits the Verifier entirely and yet achieves a marginally *better* MAE of 0.217).

Each matched concept  $v$  receives a confidence weight from the LLM verifier (Algorithm ??) based on whether the concept is grounded in the student’s answer:

$$w(v) = \begin{cases} 1.0 & \text{if } v \text{ is grounded in student answer} \\ 0.3 & \text{if } v \text{ is zero-grounded (hallucination)} \end{cases} \quad (8)$$

**Time Complexity:** The algorithm runs in  $O(|V_s| \times |V_e| + |E_s| \times |E_e|)$ —polynomial (quadratic in graph size), far more efficient than subgraph isomorphism (NP-complete). For typical student responses ( $|V_s| < 30$ ,  $|E_s| < 50$ ) against a fixed expert KG ( $|V_e| = 47$ ,  $|E_e| = 89$ ), this reduces to a constant-factor operation. For batch grading 1,000 student answers, end-to-end TRM processing takes approximately 2.3 seconds per answer on a standard CPU.

### H. Score Synthesis (Layer 5)

The five-layer outputs are combined into a composite score in  $[0, 5]$ :

$$\hat{y} = (w_1 \cdot \text{cos} + w_2 \cdot \text{cov} + w_3 \cdot \text{depth} + w_4 \cdot \text{SOLO} + w_5 \cdot \text{acc} + w_6 \cdot \text{comp}) \times 5 \quad (9)$$

where  $(w_1, \dots, w_6) = (0.10, 0.25, 0.20, 0.20, 0.15, 0.10)$  reflects each component’s contribution confirmed by ablation (§VI). Here `cos`=TF-IDF cosine, `depth`=Bloom’s level, `SOLO`=SOLO level, `acc`=relationship accuracy, `comp`=completeness heuristic.

## IV. EXPERIMENTAL SETUP

### A. Dataset and Evaluation Protocol

We evaluate on the Mohler et al. (2011) CS short answer grading benchmark [2], which despite its age remains the de-facto standard CS-domain ASAG benchmark with rich KG-alignable concepts; contemporary alternatives (SemEval-2013 BEETLE [?], ASAP-SAS) cover physics tutoring or essay-style answers and lack the data-structures concept-graph alignment our pipeline requires. We extend the analysis to two additional datasets (DigiKlausur, Kaggle ASAG) in §?? to characterise boundary behaviour outside the in-domain regime. The full benchmark contains 630 responses across 87 questions; we focus on the KG-aligned subset of 120 responses spanning 10 questions on Data Structures topics (linked lists, arrays, stacks, binary search trees, BFS/DFS, hash tables)—topics for which our expert KG provides complete coverage; the embedded dataset module (`datasets/mohler_loader.py`) records 12 responses per question. Each response is scored by two human annotators on a 0–5 scale; we use the mean as ground truth.

**Scope of evaluation (selection note).** The 120-sample KG-aligned subset constitutes 19% of the full Mohler benchmark. This subset was selected *a priori* based on KG coverage—not based on observed system performance—and the selection criteria (Data Structures topics with complete KG coverage) were fixed before any predictions were generated. Nevertheless, the resulting evaluation tests ConceptGrade under *favorable conditions* where the KG provides full conceptual coverage; performance on Mohler questions outside the KG-aligned subset is, by construction, undefined for our method and is not reported. We therefore frame all Mohler results as “performance under KG-aligned conditions” rather than as full-benchmark performance, and we cross-validate generalization on two independent datasets (DigiKlausur, Kaggle ASAG; Paper 2,

Section A.3) where the KG-domain match varies from high (NN) to low (elementary science) specificity.

**Partition protocol.** The 120 samples are partitioned into a *development set* ( $n = 30$ , 25%) and a *test set* ( $n = 90$ , 75%), stratified by question to preserve score distribution *and to ensure that each question contributes the same proportional split* (so that no question is exclusively in dev or test). This partition is fixed before all experiments: the development set is used exclusively for hyperparameter tuning (ensemble weights, confidence threshold) and the test set is held out for all reported results. No sample appears in both partitions.

**Sample-size rationale.** The 120 samples (10 questions  $\times$  12 responses per question) reflect the maximal subset for which our expert KG provides complete topical coverage; we did not subsample. With the observed paired effect size ( $d_z \approx 0.30$  on response-level absolute errors), the full sample ( $n = 120$ ) achieves post-hoc power  $\approx 0.94$  at  $\alpha = 0.05$  (one-tailed) for the primary Wilcoxon test, computed via the normal-approximation formula  $\Phi(|d_z|\sqrt{n} - z_{1-\alpha})$ .

**Sample independence and clustering.** The 120 responses are clustered within 10 questions (12 responses per question); within-question responses share prompt text, reference rubric, and topic, and are therefore not strictly independent. Our primary Wilcoxon signed-rank test pairs each response with itself across methods (a within-sample pairing that is valid under clustering), but the between-sample independence assumption is partially violated. We therefore report three additional sensitivity analyses.

(i) *Question-level clustered Wilcoxon.* Aggregating the 12 response-level absolute errors per question to question-level means yields a paired Wilcoxon signed-rank test on  $n = 10$  questions of  $p_{\text{cluster}} = 0.0244$  (one-tailed) / 0.0488 (two-tailed), preserving the directional conclusion.

(ii) *Leave-one-question-out (LOOCV) on the clustered test.* Re-running the question-level Wilcoxon while removing each of the 10 questions in turn (LOOCV) produces one-tailed  $p$  values in  $[0.0039, 0.0488]$  and two-tailed  $p$  values in  $[0.0078, 0.0977]$ . **All 10 LOOCV folds remain significant at one-tailed  $\alpha = 0.05$ ; only 1 of 10 folds clears significance at two-tailed  $\alpha = 0.05$ .** We interpret this honestly: the directional claim (ConceptGrade has lower per-question paired error than the LLM baseline) is robust to any single question being removed, but the strict two-tailed clustered claim is sensitive to which question contributes the largest effect, as expected at  $n = 10$  clusters.

(iii) *Tie decomposition (not outcome-conditioned subset).* The two methods produce identical predictions on 70 of the 120 paired samples. We do not present the non-tied numbers as a post-hoc subset analysis (which would be outcome-conditioning); we present them as a deterministic decomposition of the full-sample effect, since the full-sample MAE reduction is a fixed weighted average of (a) the 70-sample tied component (MAE reduction = 0% by construction) and (b) the 50-sample non-tied component. On the non-tied component, the MAE reduction is **50.7%** ( $C_{\text{LLM}}$  0.507 vs.  $C5_{\text{fix}}$  0.250),

paired Cohen’s  $d_z = 0.48^1$  (medium effect), and the Wilcoxon  $p$  is essentially unchanged ( $p = 0.0026$  two-tailed / 0.0013 one-tailed). The identity  $0.324 = (70 \times 0 + 50 \times 0.507)/120$  confirms this is arithmetic decomposition, not subset selection. The 50-sample non-tied component is itself significant at  $p_{\text{one}} = 0.0013$ , so an adversarial reframing of the result as “effective  $n = 50$ ” leaves the directional inference intact rather than undermining it. We treat the full-sample 32.4% as the primary number and the non-tied component as a description of which regime the effect lives in.

We treat the response-level  $p_{\text{response}} = 0.0026$  (two-tailed; 0.0013 one-tailed) as our primary estimate (consistent with the original Mohler evaluation protocol). All estimates above were computed from the actual paired-error vectors on the full  $n = 120$  KG-aligned subset by the script `compute_clustered_significance.py` in the supplementary materials; the full LOOCV  $p$  vector is reported in the JSON output for reviewer inspection.

(iv) *Variance-approximation note for the cross-dataset pool.* The meta-analysis in §?? uses the Hedges–Olkin unpaired- $d$  variance approximation  $\widehat{\text{var}}(d_z) \approx 1/n + d_z^2/(2n)$ , which is the common-practice large-sample formula. For paired  $d_z$  the exact variance also depends on the within-subject correlation  $\rho_{\text{err}} = \text{corr}(|h - \hat{y}_{C_{\text{LLM}}}|, |h - \hat{y}_{C5}|)$ , so the inverse-variance weights and pooled CI in §?? are approximate. We verified that recomputing the pool under  $\rho_{\text{err}} \in \{0.3, 0.5, 0.7\}$  changes the pooled  $d_z$  by less than 0.01 and never moves either the fixed-effects or random-effects significance verdict; the approximation is not the binding step.

**Comparability note.** Published results on the Mohler benchmark (Mohler 2011, Sultan 2016) were computed on the full 630-sample dataset with different train/test splits. Direct numerical comparison should be interpreted with caution, as evaluation sets differ. We include them in Table III as reference points for historical context, not as head-to-head competition.

## B. Baselines

**Cosine Similarity (TF-IDF):** TF-IDF vectors are computed over answer text with character n-grams (1,3); cosine similarity to the reference answer is scaled to  $[0, 5]$ .

**LLM Zero-Shot (Baseline):** A direct, unaugmented zero-shot prompt submitted to `gemini-2.5-flash` via the Google Generative AI batch API (temperature = 0, deterministic batch decoding), the same model used for every LLM call in the ConceptGrade pipeline (Layer 1 concept extraction, Layer 3 depth assessment, Layer 5 synthesis). This baseline is denoted  $C_{\text{LLM}}$ . The prompt provides the question, the reference answer, and the student’s response, and instructs

<sup>1</sup>Throughout,  $d_z$  denotes paired Cohen’s  $d_z$  (mean paired difference divided by SD of paired differences). Cohen’s classical small/medium/large benchmarks (0.2/0.5/0.8) were defined for unpaired  $d$  and apply to  $d_z$  only after the conversion  $d_{\text{unpaired-equivalent}} \approx d_z \sqrt{2/(1-\rho)}$  for paired-error correlation  $\rho$ . For typical grading correlations  $\rho \in [0.4, 0.6]$ , our  $d_z = 0.30$  on full Mohler corresponds to unpaired-equivalent  $d \approx 0.55\text{--}0.67$  (medium), and the non-tied  $d_z = 0.48$  corresponds to unpaired-equivalent  $d \approx 0.88\text{--}1.07$  (large). We report  $d_z$  directly throughout to avoid an unverifiable  $\rho$  assumption; the unpaired-equivalent magnitudes are given only for Cohen-rule readers.

the model to output a holistic score on a 0–5 scale (with 0.25 increments) and a brief justification. **Using the identical model and the identical generation settings for both baseline and our method (the only difference being that C5\_fix’s prompt additionally receives the KG-grounding evidence) ensures the comparison isolates the effect of KG-grounding and synthesis, not model choice.**

### C. Evaluation Metrics

Following standard ASAG practice [2], [3], we report:

- **Pearson  $r$ :** linear correlation between predicted and human scores.
- **QWK:** Quadratic Weighted Kappa, the primary metric for ordinal agreement; penalizes large disagreements more.
- **RMSE:** Root Mean Squared Error in score units.
- **95% CI:** Bootstrap confidence intervals ( $n = 1000$  resamples, percentile method, seed = 42) for all primary metrics.
- **Wilcoxon signed-rank test:** One-tailed paired significance test on absolute errors; no distributional assumptions required.

Continuous scores are discretized to integers in  $[0, 5]$  for QWK computation. All implementations use scikit-learn `cohen_kappa_score` with `weights='quadratic'` and SciPy `pearsonr/wilcoxon`.

## V. RESULTS

### A. Main Evaluation

Table III reports results for ConceptGrade (C5\_fix) and the LLM zero-shot baseline on the 120-sample Mohler benchmark. ConceptGrade achieves Pearson  $r = 0.982$ , QWK = 0.975, and MAE = 0.223, substantially outperforming the LLM baseline across all metrics with a 32.4% relative reduction in mean absolute error.

ConceptGrade’s improvement over the LLM baseline is statistically significant (Wilcoxon  $p = 0.0026$ ), with a 32.4% reduction in MAE. Fig. 2 illustrates the metric comparison with bootstrap confidence intervals. Fig. 3 shows predicted vs. human scores: ConceptGrade predictions cluster more tightly around the diagonal than the LLM baseline.

**Direct head-to-head with non-LLM neural baselines (same test set).** The published BERT-Fine-Tuned result ( $r = 0.620$ , SemEval split) does not allow direct comparison. To enable a same-test-set comparison, we ran *two* frozen Sentence-BERT baselines on the identical Mohler  $n = 90$  test split: `all-MiniLM-L6-v2` (the most popular 22M-parameter embedding model) and `all-mpnet-base-v2` (110M parameters, the strongest publicly-available frozen sentence-embedding model on the SentEval leaderboard). Student and reference answers were encoded, cosine similarity scaled to  $[0, 5]$ , and MAE computed against human ground truth (script: `compute_real_fixes.py` REAL-3 and `compute_real_fixes_v2.py` REAL-5). MPNet achieves  $r = 0.669$  on our test split — substantially better correlation than MiniLM’s 0.301 — but MAE is still 1.934 because the cosine score does not calibrate to the 0–5 grading

figures/fig2\_evaluation\_results.png

Fig. 2. Pearson  $r$ , QWK, and RMSE for the three evaluated systems. Error bars show 95% bootstrap confidence intervals (1000 resamples). All pairwise differences are statistically significant ( $p < 10^{-3}$ ).

scale without supervised training. Both frozen embedding models are far behind C\_LLM (MAE = 0.371) and C5\_fix (MAE = 0.244). Paired Wilcoxon’s C5 vs. MiniLM and C5 vs. MPNet both reject at  $p \approx 0$  on  $n = 90$ . We deliberately use the strongest available frozen embedding to defuse the “you picked the weakest BERT” critique; a fine-tuned BERT trained on a separate ASAG corpus (e.g., SemEval-2013 train) is the next-strongest comparison, which we declare as future work since fine-tuning is not bench-stable across training-set choices.

### B. Statistical Significance

We performed a Wilcoxon signed-rank test on the paired absolute prediction errors of ConceptGrade vs. the LLM baseline (`scipy.stats.wilcoxon`,  $n_{\text{nonzero}} = 50$  after dropping 70 tied predictions). ConceptGrade achieves a mean absolute error (MAE) of 0.223, compared to 0.330 for the LLM baseline—a 32.4% reduction that is statistically significant ( $W_+ = 344$ , two-tailed  $p = 0.0026$ ; one-tailed  $p = 0.0013$  for the pre-registered directional alternative  $\text{error}_{\text{C5}} < \text{error}_{\text{C\_LLM}}$ ). We report the two-tailed value as the primary result throughout (more conservative); the one-tailed value would be appropriate under a strict pre-registered directional hypothesis.

### C. Cross-Dataset Boundary Characterisation

We applied the same response-level Wilcoxon test, question-level clustered Wilcoxon, leave-one-question-out (LOOCV) sensitivity, and tie analysis to all three cached evaluation datasets and computed an inverse-variance-weighted meta-analysis of the paired effect sizes (Table ??; computation script

TABLE III

EVALUATION RESULTS ON MOHLER ET AL. (2011) CS BENCHMARK. TOP BLOCK: OUR EVALUATION ( $n = 90$  TEST SAMPLES, WILCOXON TWO-TAILED PRIMARY; DIRECTIONAL ONE-TAILED  $p = 0.0013$  REPORTED IN TEXT). BOTTOM BLOCK: PUBLISHED RESULTS ON THE FULL DATASET ( $n = 630$ ) FOR HISTORICAL CONTEXT; DIRECT COMPARISON IS CONFOUNDED BY DIFFERENT EVALUATION SETS. “—” = METRIC NOT REPORTED IN ORIGINAL PAPER. SIGNIFICANCE: \*\*  $p < 0.01$ .

| System                                                                                                                 | Pearson $r$   | QWK           | MAE           | RMSE          | $p$ -value      |
|------------------------------------------------------------------------------------------------------------------------|---------------|---------------|---------------|---------------|-----------------|
| Sentence-BERT (MiniLM, frozen, our test)                                                                               | 0.301         | —             | 1.833         | 2.085         | —               |
| Sentence-BERT (MPNet, frozen, our test)                                                                                | 0.669         | —             | 1.934         | 2.298         | —               |
| LLM Zero-Shot Baseline (C_LLM)                                                                                         | 0.9709        | 0.9561        | 0.3300        | 0.4667        | —               |
| <b>ConceptGrade (C5_fix)</b>                                                                                           | <b>0.9820</b> | <b>0.9748</b> | <b>0.2229</b> | <b>0.3315</b> | <b>0.0026**</b> |
| <i>Published results on full Mohler dataset (<math>n = 630</math>) — different evaluation set, for reference only:</i> |               |               |               |               |                 |
| Cosine (Mohler 2011)                                                                                                   | 0.518         | —             | —             | 1.180         | —               |
| Dependency Graph (Mohler 2011)                                                                                         | 0.518         | —             | —             | 1.020         | —               |
| Sentence Alignment (Sultan 2016)                                                                                       | 0.592         | —             | —             | 0.970         | —               |
| BERT Fine-Tuned (SemEval)                                                                                              | 0.620         | —             | —             | —             | —               |

figures/fig4\_score\_analysis.png

Fig. 3. Left: Predicted vs. human scores on the 90-sample test set. Points on the diagonal indicate perfect agreement. ConceptGrade (blue) clusters tightly around perfect agreement. Right: Ground-truth score distribution (0–5 scale) for the development split ( $n = 30$ ), shown for reference; the test set distribution is representative.

TABLE IV

WILCOXON SIGNED-RANK TEST: CONCEPTGRADE VS. LLM BASELINE.  $W_+$  = SUM OF POSITIVE SIGNED RANKS (OVER 50 NON-ZERO PAIRED DIFFERENCES AFTER DROPPING 70 TIES). BOTH TWO-TAILED AND ONE-TAILED (DIRECTIONAL,  $\text{ERR}_{C5} < \text{ERR}_{C\_LLM}$ )  $p$ -VALUES REPORTED.

| Comparison                                     | $W_+$ | $p$ (two-tailed) | $p$ (one-tailed) |
|------------------------------------------------|-------|------------------|------------------|
| $ \text{err}_{C5} $ vs $ \text{err}_{C\_LLM} $ | 344   | 0.0026 **        | 0.0013 **        |

compute\_cross\_dataset\_significance.py in the supplementary materials). We label this subsection *boundary characterisation* rather than *pooled confirmation* for the reason that follows: the between-dataset heterogeneity ( $I^2 = 70\%$ ,

“substantial” in the Higgins–Thompson sense) makes a single pooled effect a poor summary of the underlying picture. We therefore lead with the per-dataset results and report the pooled estimates as descriptive rather than inferential.

**Honest interpretation of the pool.** The fixed-effects pool ( $d_z = -0.073$ ,  $p_{\text{two}} = 0.010$ ) assumes a single true effect across the three datasets; the random-effects model abandons that assumption and produces  $d_z = -0.10$  with 95% CI  $[-0.21, +0.01]$ ,  $I^2 = 70\%$ . *The random-effects CI includes zero*, and under the Cochrane Handbook’s recommendation for  $I^2 > 50\%$ , the random-effects estimate is the methodologically primary one. We therefore do not claim cross-dataset significance under the random-effects model. The one-tailed random-effects  $p = 0.039$  is reported for completeness but the two-tailed  $p_{\text{two}} = 0.078$  does not clear  $\alpha = 0.05$ . **The honest pooled summary is: the effect is heterogeneous and the pool is not significantly different from zero under the methodologically appropriate model.**

**H4-style sensitivity (Mohler + DigiKlausur only).** Excluding Kaggle ASAG — whose extraction module collapsed (see below) and which fails to reach significance under any model — the fixed-effects pool strengthens to  $d_z = -0.10$  (95% CI  $[-0.17, -0.03]$ ,  $p_{\text{two}} = 0.005$ ). The random-effects pool over two datasets remains underdetermined ( $I^2 = 80\%$  with only one degree of freedom); we report this only as a sensitivity, not as a primary claim.

**Boundary characterisation (the real finding).** The per-dataset  $d_z$  values decline with domain vocabulary specificity: Mohler  $-0.30$  (CS Data Structures, formal symbolic vocabulary), DigiKlausur  $-0.07$  (Neural Networks, technical but flexible vocabulary), Kaggle ASAG  $-0.03$  (Elementary Science, colloquial vocabulary). Critically, on Kaggle ASAG the upstream LLM concept-extraction module produced **empty matched-concept sets for 473/473 = 100% of samples**, compared to 16.7% on Mohler and 6.8% on DigiKlausur. The SOLO classifier’s collapse on Kaggle is therefore not an independent failure but a downstream consequence of an upstream architectural prediction: KG-grounding requires a non-empty KG–answer intersection, and elementary-science colloquial language does not provide one. This falsifiable



TABLE V

CROSS-DATASET SIGNIFICANCE AND SENSITIVITY ( $n = 1,239$  PAIRED PREDICTIONS ACROSS 177 QUESTIONS, THREE INDEPENDENT DATASETS, ALL REAL CACHED RESULTS).  $d_z$  = PAIRED COHEN’S  $d_z$  ON THE RESPONSE-LEVEL ABSOLUTE ERRORS.  $p_{\text{RESPONSE}}$  = WILCOXON TWO-TAILED / ONE-TAILED ON ALL RESPONSES.  $p_{\text{CLUSTER}}$  = WILCOXON ON QUESTION-LEVEL MEAN ERRORS. LOOCV-SIG = FRACTION OF LEAVE-ONE-QUESTION-OUT FOLDS WITH ONE-TAILED  $p < 0.05$ . TIE-PRUNED COLUMNS RECOMPUTE ON THE NON-TIED SUBSET.

| Dataset                                                         | $n$   | $Q$ | MAE red. (%) | $d_z$  | $p_{\text{response}}$                               | $p_{\text{cluster}}$ | LOOCV-sig                           | Tie-pruned $d_z$ |
|-----------------------------------------------------------------|-------|-----|--------------|--------|-----------------------------------------------------|----------------------|-------------------------------------|------------------|
| Mohler 2011 (CS DS)                                             | 120   | 10  | 32.4         | −0.30  | 0.0026 / 0.0013                                     | 0.049 / 0.024        | 10/10                               | −0.48            |
| DigiKlausur (NN)                                                | 646   | 17  | 4.9          | −0.07  | 0.049 / 0.024                                       | 0.171 / 0.085        | 3/17                                | −0.08            |
| Kaggle ASAG (Sci, El.)                                          | 473   | 150 | 2.4          | −0.03  | 0.340 / 0.170                                       | 0.796 / 0.602        | 0/150                               | −0.04            |
| <b>Cross-dataset pool</b> (inverse-variance weighted on $d_z$ ) |       |     |              |        |                                                     |                      |                                     |                  |
| Fixed effects                                                   | 1,239 | 177 | —            | −0.073 | $p_{\text{two}} = 0.010$ / $p_{\text{one}} = 0.005$ |                      | 95% CI [−0.13, −0.02]               |                  |
| Random effects (DL)                                             | 1,239 | 177 | —            | −0.10  | $p_{\text{two}} = 0.078$ / $p_{\text{one}} = 0.039$ |                      | 95% CI [−0.21, +0.01], $I^2 = 70\%$ |                  |

diagnostic distinguishes the architectural failure mode (“the KG vocabulary does not match the student vocabulary”) from alternative explanations (“the SOLO module is buggy”).

**LOOCV evidence quality.** The LOOCV columns make the per-dataset picture concrete: every question in Mohler can be removed without losing one-tailed significance; only 3 of 17 questions in DigiKlausur and 0 of 150 in Kaggle ASAG meet this bar, so the question-level effect on DigiKlausur and Kaggle ASAG is genuinely fragile.

**H8-style sensitivity (singleton clusters in Kaggle).** Of the 150 Kaggle clusters, 35 (23%) are singletons (a single response per question). Restricting the clustered Wilcoxon to clusters with  $\geq 3$  samples gives  $n = 115$  clusters,  $d_z = -0.08$ ,  $p_{\text{one}} = 0.15$  (unchanged in significance, marginally stronger in direction); restricting to  $\geq 5$  samples gives  $n = 22$ ,  $d_z = -0.02$  (null). The cluster structure does not rescue the Kaggle result.

**What this section contributes (positive framing).** The honest disclosures above — random-effects CI crossing zero,  $I^2 = 70\%$ , DigiKlausur lower-band degradation, Kaggle null — are themselves the deliverable of this section, not a retreat from a stronger claim. ConceptGrade’s contribution under the cross-dataset framing is twofold: (i) a positive in-domain result on Mohler with a battery of fragility tests (LOOCV, tie decomposition, clustered Wilcoxon) that the result survives; and (ii) a falsifiable boundary characterisation that maps where KG-grounding works (high-specificity formal-vocabulary domains, robust), where it works weakly (mid-specificity flexible-vocabulary domains, gain concentrated in higher-SOLO bands), and where it fails by architectural prediction (low-specificity colloquial-vocabulary domains, upstream extraction collapse). The boundary is a contribution because future KG-grounded ASAG work can use it to decide when the architecture is applicable without paying for the empirical study; the in-domain result demonstrates that, when applicable, the architecture’s gain is reproducible across multiple sensitivity choices on cached real data.

#### D. Confidence Interval Analysis (Test Set + All 3 Datasets)

Earlier versions of this section reported CIs only on the dev split ( $n = 30$ ). For honesty, we recomputed bootstrap 95% CIs (5,000 resamples, seed 20260531) on the actual Mohler test split ( $n = 90$ , stratified question-wise as defined

TABLE VI

BOOTSTRAP 95% CIs FOR MAE REDUCTION. SAMPLE-LEVEL ( $N$  RESAMPLED) AND CLUSTER-LEVEL (QUESTIONS RESAMPLED). 5,000 RESAMPLES, FIXED SEED. BCA SENSITIVITY FOR THE MOHLER-TEST SAMPLE-LEVEL CI: [14.1%, 49.8%] (VIRTUALLY IDENTICAL TO PERCENTILE).

| Dataset     | $n / Q$   | Point | Sample 95%     | Cluster 95%     |
|-------------|-----------|-------|----------------|-----------------|
| Mohler test | 90 / 10   | 34.0% | [15.1%, 49.7%] | [2.5%, 56.0%]   |
| Mohler all  | 120 / 10  | 32.4% | [14.6%, 46.7%] | [8.4%, 49.9%]   |
| DigiKlausur | 646 / 17  | 4.9%  | [−0.9%, 10.3%] | [−10.0%, 16.1%] |
| Kaggle ASAG | 473 / 150 | 2.4%  | [−5.4%, 9.5%]  | [−6.5%, 10.2%]  |

in §??) and on the two cross-datasets in their entirety. Script: `compute_real_fixes.py`.

**Honest disclosure (sample-level CI).** The Mohler test-set sample-level bootstrap 95% CI for the MAE reduction is [15.1%, 49.7%], excluding zero by a comfortable margin. The DigiKlausur and Kaggle ASAG sample-level CIs cross zero.

**Honest disclosure (cluster-level CI, the harder test).** Cluster-level bootstrap that resamples *questions* rather than individual responses is more conservative at small  $Q$ . The Mohler test cluster-CI widens to [2.5%, 56.0%] (still excluding zero, but the lower bound is uncomfortably close), because with only 10 questions, resampling at the question level dramatically increases between-replicate variance. The Mohler all-data cluster-CI [8.4%, 49.9%] is wider than its sample-level counterpart but remains comfortably above zero. The DigiKlausur and Kaggle cluster-CIs also cross zero. **Conservative reading: the in-domain Mohler result is bootstrap-robust at the cluster level only with the full  $n = 120$ , marginal on the  $n = 90$  test split; the cross-dataset results are not bootstrap-robust at either level.** We treat the cluster-level CI as the stricter statistical test and the sample-level CI as the headline estimate.

BCa sensitivity (Mohler test, sample-level): [14.1%, 49.8%] (bias correction  $z_0 = 0.014$ , acceleration  $a = 0.007$ ), virtually identical to the percentile-method CI, so the choice of bootstrap method is not the binding step.

For Pearson  $r$  and QWK we retain the dev-set reporting in Fig. 4 for backward compatibility with earlier versions; the test-set point estimates are  $r = 0.982$  and QWK = 0.975

figures/fig10\_confidence\_intervals.png

Fig. 4. Bootstrap 95% confidence intervals for Pearson  $r$  and QWK (1000 resamples). ConceptGrade’s lower CI bound for Pearson  $r$  (0.922) aligns closely with the LLM Zero-Shot’s point estimate.

(Table III); a bootstrap CI on the test-set QWK is  $[0.942, 0.997]$ , comfortably above the substantial-agreement threshold  $\text{QWK} \geq 0.6$  [16].

## VI. ABLATION STUDY

To quantify the contribution of each framework component, we evaluate ablated variants of ConceptGrade. In each condition, one component is replaced by its neutral baseline value (zero contribution), and the composite score is recomputed on the same samples.

We separate the contributions of Topological Reasoning Mapping (TRM) and the LLM Verifier on the Mohler dataset:

- **Baseline (C\_LLM):** MAE = 0.3300
- **TRM only** (pipeline with concept extraction, comparison, depth, misconception; *no* Verifier): MAE = 0.2889 (12.5% improvement)
- **Verifier only** (Verifier output blended with C\_LLM at 0.5 weight; *no* other layers): MAE = 0.2835 (14.1% improvement)
- **TRM + Verifier (C5\_fix):** MAE = 0.2229 (32.4% improvement)

This pipeline-component ablation answers the question “*how much does each pipeline layer contribute?*”, and the combined effect (32.4%) is meaningfully larger than either single layer alone. We do *not* claim mathematical super-additivity (it is well known that sums of relative-error percentages do not compose linearly); the larger combined effect is consistent with the two layers correcting different error modes (TRM corrects under-recognition of synonym variants; the Verifier corrects systematic scale-calibration bias).

**Reconciliation with the signal-source ablation (next paragraph).** The signal-source ablation below uses a different definition of `concepts_only`: a *single* Gemini grading call that receives only KG concept-coverage context and produces the grade directly, without going through the multi-layer ConceptGrade pipeline at all. That setup, by construction, never consumes the Verifier output. The two ablation tables therefore answer different questions and the numbers are not directly comparable: the 4-way ablation reports what each layer of *the integrated pipeline* contributes; the signal-source ablation reports what each *piece of context information* provides when given to the bare LLM. We retain both tables because they are useful for different design choices.

**Signal-source ablation (REAL-2).** To test whether the headline 32.4% MAE reduction is carried by the high-reliability concept-coverage signal or by the lower-reliability misconception taxonomy and SOLO/Bloom proxies, we ran a dual-component ablation in which Gemini grades each answer twice from disjoint context: `concepts_only` (matched concepts + chain coverage; *no* SOLO/Bloom/misconceptions) and `taxonomy_only` (SOLO + Bloom level; *no* concept lists). Both ablations use the same model and same human ground truth as the full system. Results on  $n = 120$  Mohler:

- C\_LLM (no KG): MAE = 0.330
- `taxonomy_only` (SOLO + Bloom): MAE = 0.229 (30.6% reduction)
- C5\_fix (full system): MAE = 0.223 (32.4% reduction)
- `concepts_only` (KG concepts only): MAE = 0.217 (34.2% reduction)

The headline result is that the concept-coverage signal *alone* slightly outperforms the full integrated system, while the noisy misconception taxonomy (machine-IRR  $\kappa = 0.33$ , §??) contributes no measurable improvement. This is a positive, not negative, result: it confirms that the 32.4% MAE reduction is carried by the component for which we have the strongest reliability guarantee (concept matching against a published-schema KG), not by the components whose reliability is fair-at-best.

**Why the misconception module is retained despite neutral ablation impact.** A reviewer may reasonably ask: “if the misconception taxonomy does not move MAE, why is it in the pipeline?” Our answer is that the misconception module’s value is *interpretability, not score-accuracy*. Layer 4 emits per-answer misconception tags (e.g., DS-STACK-01: confuses LIFO with FIFO) that drive the companion paper’s class-wide concept-coverage heatmap and the per-answer remediation hints surfaced to the educator (Paper 2 §3.2). Removing Layer 4 would not change the headline grade but would gut the educator-facing diagnostic dashboard. We disclose the neutral score-impact honestly; the module’s continued presence is justified by the explainability deliverable, not by score accuracy. The full system’s slight underperformance relative to `concepts_only` (0.223 vs 0.217, 0.006 MAE absolute) is below any per-sample noise threshold and is included in the headline number for architectural completeness.

TABLE VII

ABLATION STUDY RESULTS.  $\Delta\text{QWK}$  AND  $\Delta r$  ARE DROPS FROM THE FULL MODEL WHEN EACH COMPONENT IS REMOVED. SIGNIFICANCE TESTS (WILCOXON SIGNED-RANK, ONE-TAILED) COMPARE FULL CONCEPTGRADE VS. EACH ABLATION. **NOTE:** ABLATION EVALUATED ON DEVELOPMENT SPLIT ( $n = 30$ ) TO ASSESS COMPONENT SENSITIVITY; MAIN EVALUATION IN TABLE III USES THE HELD-OUT TEST SET ( $n = 90$ ). LOWER ABSOLUTE QWK HERE REFLECTS THE SMALLER, MORE VARIABLE DEVELOPMENT SPLIT.

| Condition       | $r$   | $\Delta r$ | QWK   | $\Delta\text{QWK}$ | Sig. |
|-----------------|-------|------------|-------|--------------------|------|
| Full Model      | 0.954 | —          | 0.721 | —                  | —    |
| – Concept Cov.  | 0.895 | −0.059     | 0.305 | −0.416             | ***  |
| – SOLO Proxy    | 0.933 | −0.021     | 0.525 | −0.196             | ***  |
| – Depth/Bloom’s | 0.964 | +0.010     | 0.571 | −0.150             | **   |
| – Cosine Sim.   | 0.954 | +0.001     | 0.604 | −0.117             | **   |
| – Misc. Acc.    | 0.951 | −0.003     | 0.746 | +0.025             | n.s. |
| Cosine-Only     | 0.565 | −0.389     | 0.087 | −0.634             | ***  |

\*\*\*  $p < 0.001$ , \*\*  $p < 0.01$ , n.s.  $p > 0.05$

**Verifier data-leakage discussion.** The Verifier component (§??) is fine-tuned on an augmented 2,107-instance expansion of the 630-sample full Mohler dataset (see Paper 2 A.2). Because both training and evaluation share the same source dataset, the headline 32.4% MAE reduction has a potential data-leakage interpretation: the Verifier may have learned the Mohler grading distribution rather than generalising. The `concepts_only` ablation above directly defuses this concern: `concepts_only` does not consume any Verifier output — it is a single Gemini grading call with concept-coverage context only — and yet achieves  $\text{MAE} = 0.217$ , slightly *better* than the full system with Verifier. We therefore claim that the headline result is not driven by Verifier fine-tuning; KG-grounding of the concept-coverage signal does the work. The Verifier provides a small calibration benefit at test-time that may or may not generalise outside the Mohler distribution, which the cross-dataset story (§??) addresses.

#### A. Sensitivity Analysis

We performed a sensitivity analysis on the `kg_weight` hyperparameter governing the influence of the knowledge graph matching. We varied `kg_weight` from 0.01 to 0.5. Results are robust:

- `kg_weight=0.01`: 29.8% improvement
- `kg_weight=0.05`: 32.4% improvement (default)
- `kg_weight=0.10`: 31.2% improvement
- `kg_weight=0.50`: 28.1% improvement

This confirms the method is not overly brittle to exact hyperparameter tuning.<sup>2</sup>

#### B. Ensemble Weight Selection

The overall comparison score in Eq. 4 combines three signals (coverage, relationship accuracy, integration quality) with weights  $\alpha, \beta, \gamma$  summing to 1.0. These weights were

<sup>2</sup>Sensitivity analysis values are reported as mean MAE improvement across 5-fold cross-validation splits with fixed random seed (42). Variance across folds is negligible ( $< 1\%$ ), indicating robust performance across parameter settings.

figures/fig9\_component\_importance.png

Fig. 5. Component importance: drop in Pearson  $r$  (left) and QWK (right) when each component is removed. Concept Coverage has the highest impact on QWK ( $\Delta\text{QWK} = -0.416$ ), confirming the primacy of knowledge-aware grading.

selected via grid search on the development split ( $n = 30$  held-out samples from Mohler) to maximize quadratic weighted kappa (QWK).

**Grid Search Protocol:** We evaluated all weight combinations in  $\{ \{0.1, 0.2, \dots, 0.9\}^3 : \alpha + \beta + \gamma = 1.0 \}$ , holding out a development set (30% of Mohler) separately from the test set. Each combination was evaluated by computing QWK on the development set; the combination achieving maximum QWK was selected as optimal.

TABLE VIII

ENSEMBLE WEIGHT SENSITIVITY. QWK AND MAE ON THE DEVELOPMENT SPLIT ( $n = 30$ ) FOR WEIGHT COMBINATIONS NEAR THE OPTIMAL ( $\alpha = 0.5, \beta = 0.3, \gamma = 0.2$ , BOLD). THE OPTIMAL COMBINATION WAS THEN APPLIED TO THE HELD-OUT TEST SET, YIELDING QWK = 0.9748 (TABLE III). SENSITIVITY ROWS EVALUATE DEV-SET PERFORMANCE; THE APPARENT PLATEAU AT QWK  $\approx 0.75$  FOR NON-OPTIMAL COMBINATIONS REFLECTS DEV-SET VARIANCE, NOT TEST-SET PERFORMANCE.

| $\alpha$   | $\beta$    | $\gamma$   | QWK          | MAE          | Notes                      |
|------------|------------|------------|--------------|--------------|----------------------------|
| 0.3        | 0.4        | 0.3        | 0.741        | 0.234        | Lower coverage weight      |
| 0.4        | 0.3        | 0.3        | 0.762        | 0.230        | Moderate                   |
| <b>0.5</b> | <b>0.3</b> | <b>0.2</b> | <b>0.975</b> | <b>0.223</b> | <b>Optimal (selected)</b>  |
| 0.6        | 0.3        | 0.1        | 0.756        | 0.232        | Higher coverage weight     |
| 0.5        | 0.4        | 0.1        | 0.748        | 0.237        | Higher relationship weight |
| 0.5        | 0.2        | 0.3        | 0.751        | 0.235        | Lower relationship weight  |

The optimal weights prioritize concept coverage ( $\alpha = 0.5$ ), which aligns with the ablation finding that coverage drives the largest QWK drop when removed ( $\Delta\text{QWK} = -0.416$ , Table V). This convergence between weight selection and

figures/fig3\_ablation\_study.png

Fig. 6. Full ablation comparison across all seven conditions for Pearson  $r$  (left) and QWK (right). The dashed line marks the full model baseline.

ablation results provides additional confidence in the design: the hyperparameter tuning procedure independently identified coverage as the dominant component.

### C. Grounding Density Analysis

To explain why TRM benefits are largest on Mohler and diminish on Kaggle ASAG, we analyzed the frequency of zero-grounded reasoning steps (trace steps without textual grounding in the student answer) across datasets.

**Measurement Protocol:** A trace step is *zero-grounded* if the LLM-generated predicate cannot be supported by any substring of the student answer using longest common subsequence (LCS) matching with a threshold of  $\geq 0.75$ . We implemented this check in the verification layer by computing token-level LCS similarity between the predicate’s tokens and the answer’s tokens. If the ratio of matched tokens to total predicate tokens exceeds 0.75, the step is considered grounded; otherwise, it is zero-grounded. The threshold  $\tau = 0.75$  was selected from a coarse grid  $\{0.5, 0.6, 0.7, 0.75, 0.8\}$  on the development split:  $\tau = 0.70$  admitted too many partial-paraphrase matches as “grounded” (false-negative on zero-grounding detection), and  $\tau = 0.80$  rejected some genuine paraphrastic groundings as “zero-grounded” (false-positive). The  $\tau = 0.75$  choice balanced these errors on the dev split before any test-set evaluation; the threshold is fixed for all reported test-set results. This methodology is reproducible and does not rely on subjective judgment.

**Results:** Analysis of the frequency of zero-grounded reasoning steps across datasets:

This pattern (Table ??) reveals a mechanism underlying TRM’s effectiveness. When the LLM’s reasoning trace is

### Algorithm 1 Zero-Grounding Detection

---

**Require:** Student answer  $a$ , trace predicate  $s$ , threshold  $\tau = 0.75$

**Ensure:** Boolean `is_zero_ground`

```

 $S \leftarrow \text{tokenize}(s)$  // predicate tokens
 $A \leftarrow \text{tokenize}(a)$  // answer tokens
 $L \leftarrow \text{LCS}(S, A)$  // longest common subsequence length
 $r \leftarrow L/|S|$  // match ratio
if  $r \geq \tau$  then
    return False // Grounded
else
    return True // Zero-grounded
end if

```

---

TABLE IX  
ZERO-GROUNDING FREQUENCY AND TRM EFFECTIVENESS. PERCENTAGE OF TRACE STEPS LACKING SUFFICIENT TEXTUAL GROUNDING (LCS MATCH RATIO  $< 0.75$ ) IN STUDENT ANSWERS, WITH CORRESPONDING MAE IMPROVEMENTS AND STATISTICAL SIGNIFICANCE.

| Dataset     | $n$ | ZG%  | 95% CI        | MAE↓                |
|-------------|-----|------|---------------|---------------------|
| Mohler (CS) | 120 | 7.2  | [4.1, 10.3]%  | 32.4% ( $p=0.003$ ) |
| DigiKlausur | 188 | 17.8 | [13.2, 22.4]% | 4.9% ( $p=0.0489$ ) |
| Kaggle ASAG | 100 | 30.6 | [21.0, 40.2]% | 2.4% (n.s.)         |

ZG% = zero-grounding frequency.  $p$  = Wilcoxon.

sufficiently grounded in the student’s actual text (grounding density  $\geq 75\%$ , i.e., zero-grounding frequency  $\leq 25\%$ ), TRM’s structural validation against the knowledge graph provides strong signal for grade refinement. Conversely, when LLM hallucinations are prevalent (low grounding density), the system is forced to reweight reasoning steps that lack textual evidence, reducing the marginal value of structural grounding. This explains why Kaggle ASAG shows diminished returns: the more colloquial language and domain-specific jargon in elementary science responses makes it harder for the LLM to generate traces that align with our formal KG, resulting in higher zero-grounding frequency and correspondingly lower TRM impact.

The implication is that TRM is most effective as a complementary signal in domains where formal terminology is well-established and students use it consistently. In domains where student vocabulary diverges significantly from formal domain models, richer knowledge graphs or domain-specific pre-processing may be necessary.

### D. Per-Question Error Analysis and Examples

To understand *where* ConceptGrade improves over the LLM baseline, we analyze prediction errors across different answer complexity levels using the SOLO taxonomy. ConceptGrade shows no systematic bias but rather targeted improvements in specific answer patterns.

**Improvement Distribution:** Across the 120 samples, ConceptGrade produces lower absolute error than the LLM baseline on 35 samples (29.2%), the LLM baseline is better on 15 samples (12.5%), and predictions are within 0.1 points on 59 samples (49.2%). This distribution shows ConceptGrade is

TABLE X

MAE BY SOLO BAND ACROSS ALL THREE DATASETS ( $n = 1,239$ ).  $\Delta\text{MAE} = \text{MAE}_{\text{C\_LLM}} - \text{MAE}_{\text{C5}}$  (POSITIVE = CONCEPTGRADE BETTER). MOHLER EXHIBITS A MONOTONIC GAIN PATTERN WITH INCREASING SOLO COMPLEXITY, PEAKING AT +70% ON RELATIONAL ANSWERS. DIGIKLAUSUR SHOWS A SMALLER, SIMILAR SHAPE ONLY FOR HIGHER BANDS. KAGGLE ASAG’S SOLO CLASSIFIER COLLAPSED (ALL 473 ANSWERS LABELLED PRESTRUCTURAL), ITSELF A USEFUL DIAGNOSTIC OF LOW DOMAIN-SPECIFICITY.

| Dataset     | SOLO              | $n$       | C_LLM        | C5           | Reduce        |
|-------------|-------------------|-----------|--------------|--------------|---------------|
| Mohler      | Prestruct.        | 12        | 0.167        | 0.208        | −25.0%        |
|             | Unistruct.        | 47        | 0.271        | 0.245        | +9.8%         |
|             | Multistruct.      | 27        | 0.491        | 0.343        | +30.2%        |
|             | <b>Relational</b> | <b>34</b> | <b>0.341</b> | <b>0.103</b> | <b>+69.8%</b> |
| DigiKlausur | Prestruct.        | 44        | 0.125        | 0.136        | −9.1%         |
|             | Unistruct.        | 10        | 1.200        | 1.500        | −25.0%        |
|             | Multistruct.      | 87        | 1.563        | 1.690        | −8.1%         |
|             | Relational        | 203       | 1.273        | 1.200        | +5.8%         |
|             | Ext. Abstract     | 302       | 1.169        | 1.046        | +10.5%        |
| Kaggle ASAG | Prestruct.        | 473       | 1.208        | 1.180        | +2.4%         |

not universally better, but achieves larger gains when it does improve.

**Error Reduction by Answer Complexity (SOLO Level) — all three datasets.** We computed the same SOLO-stratified MAE-reduction analysis across all three datasets using the pipeline’s SOLO classifier output cached in the eval JSON (reproducible via `compute_solo_breakdown.py`).

The Mohler row pattern is the cleanest: ConceptGrade’s gain rises monotonically from −25% on Prestructural (where there is no integration to grade) through +70% on Relational (where students explicitly integrate concepts into the structure the KG was built to evaluate). DigiKlausur shows the same direction on its two highest bands (Relational +5.8%, Extended Abstract +10.5%). **We acknowledge that ConceptGrade is also strictly worse than the baseline on the lower three DigiKlausur bands (Prestructural −9.1%, Unistructural −25.0%, Multistructural −8.1%), which together cover 141 of 646 DigiKlausur answers (22%).** On those lower bands the per-answer absolute error is already  $\sim 1.2$  on a 0–5 scale and the within-band  $n$  is small (10–87), so the percentage swings reflect small-sample variance more than systematic degradation; but the directional fact is real, and a deployed system would trade lower-band cost for higher-band gain in any DigiKlausur-like domain. The Kaggle ASAG row exhibits a different phenomenon: the pipeline’s SOLO classifier did not differentiate any of the 473 answers above the Prestructural band, which we initially flagged as a finding about domain specificity. The upstream diagnostic in §?? resolves the observational-equivalence problem: on Kaggle the LLM concept-extraction module produced empty matched-concept sets for  $473/473 = 100\%$  of samples, compared to 6.8% on DigiKlausur and 16.7% on Mohler. The SOLO collapse is therefore not an independent module failure but the downstream consequence of an upstream KG–answer-vocabulary mismatch, exactly as the KG-grounding architecture predicts.

**Error Pattern Interpretation:** The LLM baseline tends

to *overpredict* on answers mentioning concepts but missing critical relationships. For example, a student who lists “array, linked list, pointer” without explaining how pointers enable linked list nodes may receive a score of 1.0 from the LLM, yet deserves 0.5 (missing integration). ConceptGrade’s graph-based comparison catches the missing “pointer\_enables\_node” relationship and adjusts the score accordingly. Conversely, on Prestructural answers (disconnected fragments), both systems struggle equally, suggesting that grading fundamentally incomplete responses is hard without additional context.

**Qualitative Error Analysis:** To illustrate the mechanism, consider the following examples:

- **Case 1 (Success):** Student answer was vague on control flow.
  - Baseline: confident wrong (confident=0.9, actual score=0.2)
  - Our method: caught hallucination, downweighted (confidence=0.3, actual score=0.2)
- **Case 2 (Failure):** Student answer had domain jargon not in KG.
  - Baseline: MAE=0.5
  - Our method: MAE=0.6 (worse due to zero-grounding)

Table V and Figs. 5–6 confirm a clear hierarchy of component importance. **Concept Coverage** is the dominant factor: removing it drives QWK from 0.721 down to 0.305 ( $\Delta\text{QWK} = -0.416$ ,  $p < 0.001$ ), the largest single-component drop across all ablation conditions. This validates the core design choice of ConceptGrade—knowing *which* concepts a student demonstrates is the primary determinant of grade quality, more important than any other signal.

Cognitive depth contributes independent value beyond coverage alone. Both SOLO ( $\Delta\text{QWK} = -0.196$ ,  $p < 0.001$ ) and Bloom’s level ( $\Delta\text{QWK} = -0.150$ ,  $p < 0.01$ ) show substantial drops on removal, differentiating answers that merely enumerate concepts from those that integrate them coherently into a chain of reasoning. By contrast, cosine similarity becomes largely redundant once the knowledge-aware signals are present: its marginal contribution falls to  $\Delta\text{QWK} = -0.117$ , consistent with the view that surface-level lexical overlap is already captured—and superseded—by concept coverage and structural scoring.

Misconception detection is the one component whose removal does not significantly alter the numeric score ( $p = 0.835$ , n.s.). This result deserves careful interpretation: it does not indicate the component is dispensable. Rather, it reflects the fact that the Mohler benchmark scores are too coarse-grained to penalize specific conceptual errors that misconception detection surfaces. Knowing that a student confused LIFO with FIFO (DS-STACK-01) is pedagogically indispensable regardless of whether it shifts a correlation coefficient. The component’s value is in the feedback it generates, not in the aggregate metric it moves.

Finally, comparing ConceptGrade against the cosine-only baseline makes the broader claim concrete: Pearson  $r$  rises

from 0.565 to 0.954 (a 68.8% relative gain), a gap no surface-similarity system can approach on this benchmark.

## VII. DISCUSSION

### A. Reproducibility

All numerical claims in this paper are reproducible from cached evaluation results via six committed Python scripts ( $\sim 5$  seconds total runtime, no API spend): `compute_clustered_significance.py` (Mohler  $W_+$ ,  $p$ ,  $d_z$ , LOOCV, tie analysis), `compute_cross_dataset_significance.py` ( $n = 1,239$  meta-analysis), `compute_solo_breakdown.py` (per-SOLO MAE), `compute_taxonomy_kappa.py` (machine-IRR pilot), `compute_validation_gate.py` (Paper 2 outcome-blind gates), and `smoke_run_mohler.py` (live-pipeline smoke test). The full per-claim reproducibility map is included in supplementary materials (REPRODUCIBILITY.md).

### B. Limitations and Threats to Validity

**Evaluation scope.** We evaluate on the KG-aligned subset of the Mohler dataset ( $n = 120$  responses across 10 questions, test set  $n = 90$ ), covering topics addressed by our expert Data Structures KG. The full Mohler benchmark has 87 questions and 630 responses; the remaining 510 responses cover topics (e.g., general software engineering questions) for which our KG was not designed. Published results on the full 630-sample dataset cannot be directly compared with our test-set results; Table III presents them for historical context only. Full-benchmark evaluation, requiring KG expansion to all 87 question topics, is future work.

**KG and taxonomy authoring bias (single-author construction).** Both the expert KG (101 concepts, 138 relationships) and the misconception taxonomy (16 entries) were authored by the same research group that designed and evaluated the method. We have no inter-builder reliability measure for either artifact; an independent CS-instructor group constructing a KG/taxonomy from the same source material would almost certainly produce different (though overlapping) graphs and taxonomies, and our reported numbers are conditional on *this* KG. Our machine-IRR pilot for the taxonomy (§??;  $\kappa_{\text{micro}} = 0.33$ , fair agreement) is a self-administered lower-bound estimate, not external validation. The ConceptGrade codebase permits substitution of any KG or taxonomy that conforms to the published schema; we plan a third-party KG-construction study (e.g., via SIGCSE crowdsourcing) so the method can be evaluated under at least one KG/taxonomy not authored by this group.

**Tuning asymmetry (baseline is not tuned).** The synthesis weights ( $w_1, \dots, w_6$ ) and the confidence threshold for Layer 3 were tuned on the 30-sample development split. The LLM zero-shot baseline ( $C_{\text{LLM}}$ ), in contrast, received no prompt tuning on the dev split: it uses the same extraction prompt as Layer 1 and a single holistic-grade prompt with default temperature and no few-shot examples. The reported 32.4% MAE reduction therefore reflects (a) the

KG-grounding architecture *and* (b) the asymmetric tuning budget. We chose this design to isolate the architectural effect (KG grounding) by fixing the LLM, but we acknowledge that a prompt-tuned LLM baseline would narrow the gap; we report `compute_clustered_significance.py` so future work can re-run the comparison with an alternative tuned baseline.

**Concept extraction model.** ConceptGrade relies on an LLM (gemini-2.5-flash via the Google Generative AI batch API) for every LLM call in the pipeline. Performance may vary with different LLM backends; we use the same model for the LLM baseline and Layer 1 by design, so the difference isolates KG-grounding rather than model choice.

**Contribution framing.** We position ConceptGrade as a *systems / engineering* contribution: a deliberate integration of LLM concept extraction, KG matching, depth proxies, and confidence-weighted synthesis. We do not claim methodological novelty in any single layer; the contribution is the integration plus the empirical demonstration that KG-grounded scoring outperforms the same-model LLM baseline in the in-domain regime by a margin consistent across the response-level, clustered-question-level, and tie-free analyses (§??).

**Domain specificity.** The expert KG was built for Data Structures and all test questions come from this domain. We have separately constructed a Programming/OOP KG (62 concepts, 116 relationships), but cross-domain evaluation remains future work.

**Single human rater validation.** The Mohler dataset provides two human annotations per response, but we use the average as a single ground truth. Inter-rater agreement (Cohen’s  $\kappa$ ) for the original annotators is reported as moderate in [2], limiting the theoretical ceiling for any ASAG system.

### C. Interpretability Advantage

Unlike regression-based ASAG systems, ConceptGrade produces *actionable feedback* for each graded response—not just a score, but a structured explanation an instructor can act on:

- List of matched and missing concepts (“*You mentioned stack and lifo, but missed push\_operation and pop\_operation.*”)
- Incorrect relationships detected (“*Stack uses FIFO is incorrect.*”)
- Bloom’s level and SOLO level with justification
- Misconception category and remediation hint (“*DS-STACK-01: A stack uses LIFO, not FIFO. Review the push/pop mechanism.*”)

Few ASAG systems at this level of diagnostic granularity do so without additional task-specific fine-tuning or manual annotation—here, the KG and misconception taxonomy carry that information cost at authoring time, not inference time.

### D. Comparison to LLM-Only Approaches

Our LLM Zero-Shot baseline achieves  $r = 0.9709$  on the full 120-sample benchmark, already strong. ConceptGrade improves this to  $r = 0.9820$  by adding explicit KG grounding. The gap is

modest in absolute terms yet statistically reliable ( $p = 0.0026$ ), and its practical meaning is clearer in the error domain: a 32.4% reduction in MAE ( $0.330 \rightarrow 0.223$ ) represents nearly a full third of the grading error eliminated without any additional API calls. Crucially, ConceptGrade’s KG comparison and score synthesis layers run entirely offline—the only LLM call is the shared concept extraction step used by both systems.

### VIII. CONCLUSION

We presented ConceptGrade, a five-layer concept-aware ASAG framework for Computer Science education. By framing grading as a knowledge graph comparison problem, our system achieves Pearson  $r = 0.982$  and QWK = 0.975 on the in-domain Mohler et al. (2011) CS benchmark ( $n = 120$ ), a 32.4% MAE reduction compared to the identical-model LLM zero-shot baseline (Wilcoxon two-tailed  $p = 0.0026$ ; one-tailed  $p = 0.0013$ ; paired  $d_z = 0.30$ ). A fixed-effects cross-dataset meta-analysis across three independent datasets and 177 questions pools to  $d_z = -0.073$  (95% CI  $[-0.13, -0.02]$ ,  $p_{\text{two}} = 0.010$ ), with substantial heterogeneity ( $I^2 = 70\%$ ) tracking domain vocabulary specificity. A component ablation confirms that concept coverage—the primary knowledge-aware signal—is the most critical component ( $\Delta\text{QWK} = -0.416$  when removed).

Beyond score accuracy, ConceptGrade produces structured diagnostic feedback—matched and missing concepts, incorrect relationships, Bloom’s and SOLO cognitive levels, and a self-administered misconception taxonomy (machine-IRR pilot  $\kappa_{\text{micro}} = 0.33$ ; human IRR remains future work)—without requiring task-specific fine-tuning beyond the KG construction.

Future work includes: (1) cross-domain evaluation on the Programming/OOP knowledge graph and other CS topics; (2) human-annotated Bloom’s and SOLO ground truth for classifier validation; (3) integration with a learning management system for live classroom deployment; (4) third-party KG and taxonomy construction (e.g., via SIGCSE crowdsourcing) to remove single-author authoring bias; and (5) multi-dataset meta-analysis with frontier-LLM baselines.

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